Software Requirements Specification

Facial Expression-Based Depression Assessment Using Mobile Vision

1. Introduction

The proposed system is a course-project prototype for estimating depression-related behavioral cues from face images or short facial videos captured through a mobile or simulated mobile camera. It combines self-report scores such as PHQ-9/PHQ-8 with computer-vision features including facial landmarks, action units, head pose and temporal expression patterns. The output is a risk-assessment summary for research and educational use, not a medical diagnosis.

Problem Statement

Manual depression screening depends heavily on interviews and self-report scales. The project addresses the need for an objective, repeatable and low-burden support tool that can analyze facial behavior from mobile-camera data and produce measurable risk indicators for academic evaluation.

Objectives

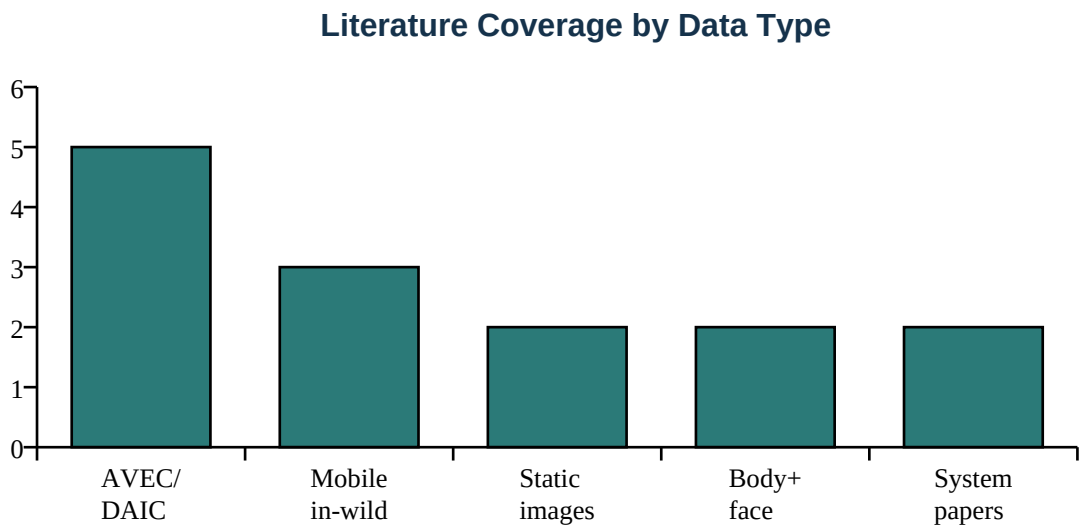
- Review existing facial-expression and mobile-sensing approaches for depression assessment.
- Implement a reproducible baseline using a public benchmark, preferably AVEC 2014 or DAIC-WOZ, with facial landmarks and temporal modeling.
- Design a mobile-vision workflow with privacy controls, quality checks, model inference and report generation.
- Evaluate the prototype using accuracy, F1-score, MAE/RMSE and ethical usability criteria.

Scope

In scope: literature survey, dataset study, preprocessing pipeline, facial feature extraction, baseline model, evaluation dashboard and SRS/design documentation. Out of scope: treatment recommendation, emergency-risk detection, production deployment and use on real users without ethics approval.

2. Literature Analysis

The reviewed literature shows three major streams: controlled benchmark videos, smartphone/in-the-wild capture, and hybrid systems that combine facial cues with passive sensors or self-report questionnaires.



No.	Paper	Year	Authors	Method and Dataset	Use for Project
1	MoodCapture	2024	Nepal et al.	In-the-wild Android front-camera images with PHQ-8 EMA; RF on 3D landmarks achieved balanced accuracy 0.60, EfficientNet 0.61.	High domain relevance; dataset private and performance modest.

No.	Paper	Year	Authors	Method and Dataset	Use for Project
2	FacePsy	2024	Islam and Bae	Open-source mobile affective sensing system using facial behavior and head gestures in naturalistic settings.	Excellent implementation reference; less useful as a public benchmark.
3	FacialPulse	2024	Wang et al.	RNN-based temporal facial landmark framework; AVEC2014/MMDA; reports 21% MAE reduction and 100% speed gain over baselines.	Strong base-paper candidate for implementation.
4	Micro-expression recognition	2023	Chen and Luo	Targets subtle facial micro-expressions for depression detection.	Useful feature idea; may be difficult without high-FPS data.
5	MoodCam	2024	Islam, Zhang and Bae	15,995 real-world smartphone interactions from 25 users; AUC 0.58-0.64 for valence and 0.60-0.63 for arousal.	Good mobile framing; small sample.
6	Faces of the Mind	2024/2025	Xu, Zhang and Zhang	11,427 adolescent facial videos; Symptom Discrepancy Index improves F1 from about 50% to about 80% after filtering heterogeneous cases.	Strong large-scale evidence; dataset access may be constrained.
7	Face alignment and scheduling	2022	Cañellas et al.	ResNet-50 streams with face alignment, augmentation, optimization and scheduling; competitive on benchmark facial-video datasets.	Good preprocessing/training baseline.
8	Fast R-CNN model	2022	Lee and Park	Conceptual chatbot/cloud system using Fast R-CNN to inspect eye/lip regions and assist depressive-disorder assessment.	Clear architecture reference; not a complete evaluated depression classifier.
9	Smartphone ML features	2022	Korean healthcare study	Mental Health Protector app collects passive sensors, survey data and facial-expression features; 106 participants.	Useful for SRS requirements beyond vision.
10	Face/body DCNN	2022	Liu et al.	Combines facial expression and body movement into behavioral depression degree; reports over 74% Pearson similarity with scales.	Good multimodal extension.
11	CCANet/FPN	2023	PLOS One	MTCNN face crop, feature pyramid, cross-channel attention; AVEC 2014; RMSE 8.65 and MAE 6.66.	Strong benchmark-paper candidate.
12	Facial image DCNN	2022	Medical Science Monitor	Static facial images; FCN, VGG, ResNet and Inception; FCN reported 98.23% accuracy.	High reported performance but likely constrained dataset.
13	Cross-corpus validation	2020	Pampouchidou et al.	Automated facial video recognition of depression/anxiety severity; evaluated generalizability including AVEC 2014.	Good validation and generalization reference.
14	ScienceDirect base link	Visible screenshot	Unknown from partial URL	Screenshot shows Elsevier article identifier S154622182005458 under project title.	Needs full citation confirmation before final submission.

3. Selected Base Papers

Primary base paper: FacialPulse [3], because it directly addresses temporal facial landmarks and offers an efficient RNN pipeline suitable for a student implementation. Secondary base paper: CCANet/FPN [10], because it uses AVEC 2014 and provides a strong image-frame baseline. Supporting base paper: MoodCapture [1], because it defines the mobile-vision problem most closely aligned with the project title.

4. Functional Requirements

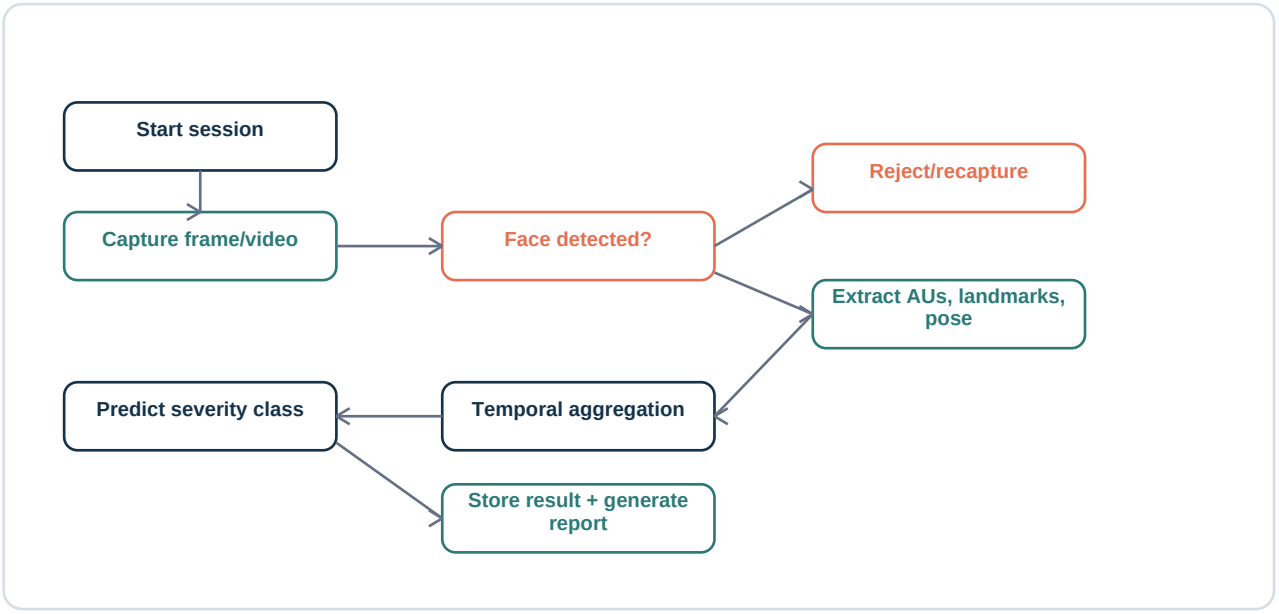
ID	Requirement	Description
FR-1	User/subject profile	Create anonymized subject/session IDs and store consent status.
FR-2	Input capture	Accept images, webcam frames or benchmark facial-video clips.

ID	Requirement	Description
FR-3	Quality validation	Reject frames with no face, severe blur, poor illumination or unsafe content.
FR-4	Feature extraction	Extract face crop, landmarks, action units, head pose and optional gaze features.
FR-5	Model inference	Predict depression class or severity score using selected baseline models.
FR-6	Evaluation	Compute accuracy, precision, recall, F1-score, MAE and RMSE as applicable.
FR-7	Report	Generate session summary with model confidence, limitations and cited dataset/source information.
FR-8	Admin view	Show dataset statistics, confusion matrix and comparative model performance.

5. Non-Functional Requirements

Category	Requirement
Privacy	Prefer on-device or local processing; do not store raw facial images unless explicitly required for coursework evidence.
Security	Use anonymized IDs, restricted folders and encrypted storage for any sensitive test data.
Usability	Keep capture steps short, visible and consent-based; provide plain-language result labels.
Performance	Process at least 10-15 fps for feature extraction on stored videos or near-real-time webcam input.
Reliability	Log preprocessing failures and allow reprocessing with fixed random seeds.
Ethics	Display a disclaimer that predictions are experimental and must not replace a qualified professional.

6. System Workflow



7. Dataset And Testing Scheme

Preferred benchmark: AVEC 2014 depression sub-challenge, because several reviewed papers report MAE/RMSE on it. Optional benchmark: DAIC-WOZ, if access is granted. For mobile context, MoodCapture and MoodCam datasets are highly relevant but appear non-public or limited; they should be cited as design evidence rather than assumed implementation data.

Dataset	Use	Source/URL	Metrics
AVEC 2014	Primary implementation benchmark	https://sspnet.eu/avec2014/	MAE, RMSE, Pearson correlation
DAIC-WOZ	Optional depression interview benchmark	https://dcapswoz.ict.usc.edu/	Accuracy, F1, MAE/RMSE
MoodCapture	Mobile design evidence	https://arxiv.org/pdf/2402.16182.pdf	Balanced accuracy, MCC, MAE
MoodCam	Mobile affect prediction evidence	https://arxiv.org/pdf/2412.12625.pdf	AUC for valence/arousal

8. Acceptance Criteria

- A literature table contains at least 6-7 detailed papers and a broader reference set.
- A working preprocessing and inference notebook/script runs on a sample video/image set.
- The project presentation includes a block diagram, workflow, dataset URL, metrics and IEEE citations.
- The final folder contains SRS, design report, PPT, literature survey sheet, paper PDFs, test data and base papers.

9. Activity Schedule

Milestone	Activity	Output
Week 1	Literature survey and base paper selection	LS sheet and top-5 paper summary
Week 2	Dataset access, preprocessing and feature extraction	Cleaned sample features
Week 3	Baseline model implementation	Training/evaluation notebook
Week 4	UI/report prototype and design refinement	Demo and SDD
Week 5	Testing, PPT and final packaging	Zipped submission folder

10. References

[1] S. Nepal et al., "MoodCapture: Depression Detection Using In-the-Wild Smartphone Images," CHI, 2024.

[2] R. Islam and S. W. Bae, "FacePsy: An Open-Source Affective Mobile Sensing System," arXiv:2406.17181, 2024.

[3] R. Wang et al., "FacialPulse: An Efficient RNN-based Depression Detection via Temporal Facial Landmarks," arXiv:2408.03499, 2024.

[4] X. Chen and T. Luo, "Catching Elusive Depression via Facial Micro-Expression Recognition," arXiv:2307.15862, 2023.

[5] R. Islam, T. Zhang, and S. W. Bae, "MoodCam: Mood Prediction Through Smartphone-Based Facial Affect Analysis in Real-World Settings," arXiv:2412.12625, 2024.

[6] X. Xu, X. Zhang, and Y. Zhang, "Faces of the Mind: Unveiling Mental Health States Through Facial Expressions in 11,427 Adolescents," arXiv:2405.20072, 2024.

[7] M. L. Cañellas, C. Á. Casado, L. Nguyen, and M. B. López, "Improving Depression Estimation from Facial Videos with Face Alignment, Training Optimization and Scheduling," arXiv:2212.06400, 2022.

[8] Y.-S. Lee and W.-H. Park, "Diagnosis of Depressive Disorder Model on Facial Expression Based on Fast R-CNN," Diagnostics, vol. 12, no. 2, Art. 317, 2022.

[9] D. Liu et al., "Measuring Depression Severity Based on Facial Expression and Body Movement Using Deep Convolutional Neural Network," Frontiers in Psychiatry, vol. 13, Art. 1017064, 2022.

[10] PLOS One, "Automatic Diagnosis of Depression Based on Attention Mechanism and Feature Pyramid Model," PLOS One, 2023, doi: 10.1371/journal.pone.0295051.

[11] "Automatic Identification of Depression Using Facial Images with Deep Convolutional Neural Network," Medical Science Monitor, 2022.

[12] A. Pampouchidou et al., "Automated Facial Video-Based Recognition of Depression and Anxiety Symptom Severity: Cross-Corpus Validation," Machine Vision and Applications, vol. 31, Art. 30, 2020.

11. Final Submission Checklist

Folder	Required Content	Status
SRS_and_Design_Report/	Updated SRS PDF and software design document PDF	Ready
PPT/	Presentation with citations, methodology, dataset URL and diagrams	To prepare
Literature_Survey/	Filled Excel sheet and at least 15 paper PDFs	To prepare
Test_Data/	Benchmark data link, sample clips/features and label notes	To prepare
Base_Papers/	FacialPulse, CCANet/FPN and MoodCapture papers	Ready to collect